A Normalized Spatial–Spectral Supervoxel Segmentation Method for Multispectral Point Cloud Data

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***Abstract*— Airborne LiDAR point cloud segmentation (PCS) is often employed as a preprocessing step for the subsequent object recognition for scene interpretation. Current segmentation methods often aim at single-wavelength LiDAR data by fully exploiting the spatial information, which makes them unsuitable for multispectral point cloud (MPC) data due to ignoring the use of spectral signatures. In this article, a normalized spatial–spectral supervoxel segmentation method is proposed for MPC data. Specifically, a normalized spectral–spatial metric is developed to construct the *k*-dimensional tree (KD tree) for MPC data clustering. Considering the uneven density distribution of MPC, an adaptive energy minimization principle based on the sum of the distance is devised to accurately select the seed points of voxels, solving the problem of undersegmentation. To reduce the cross-boundary points, the normalized spectral–spatial metric with the concave–convex judgment is extended to further optimize the edges between adjacent voxels. An important asset of our method is to segment MPC without the need for any manual annotation. Experiments on two MPC datasets show that the proposed method yields better performance compared to several comparative methods.**

***Index Terms*— Airborne LiDAR, multispectral LiDAR data, point cloud, supervoxel segmentation.**

I. Introduction

MULTISPECTRAL point cloud (MPC) data encompass both 3-D spatial and multispectral information, which vastly expands the scope of potential applications. Compared with single-wavelength point cloud data, MPC with multispectral information has potential in object detection [1], [2] and land cover classification [3], [4], [5], [6], [7]. MPC can be generated in two common ways: data fusion [8], [9], [10], [11] and multispectral LiDAR (MS-LiDAR) [12], [13], [14]. In particular, MS-LiDAR based on active remote sensing offers the advantage of being unaffected by illumination conditions and devoid of fusion errors [15].

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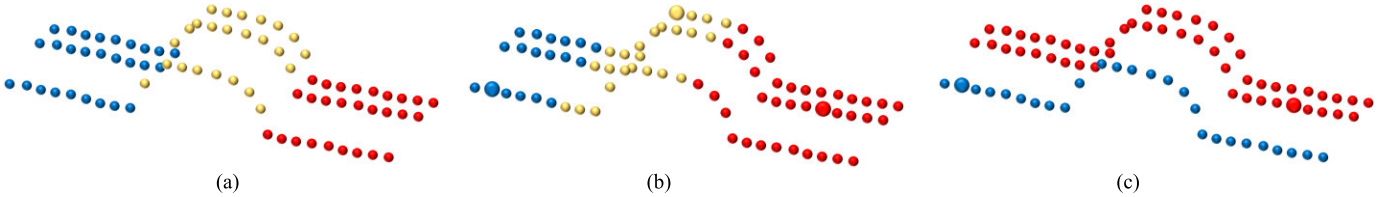
In processing large-scale point cloud data, point cloud segmentation (PCS) is often employed to improve processing efficiency as a preprocessing step [3], [11], [21] for the subsequent object recognition for scene interpretation, which can minimize the computational burden of data. Fast and accurate segmentation methods are of paramount importance for effective preprocessing in tasks [7], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26].

PCS methods draw on spatial geometry (coordinates, normal vectors, and geometric features) and intensity information, broadly categorized as edge-based [27], [28], [29], [30], region growing [31], [32], [33], [34], and model fitting [35], [36], [37], [38] techniques. Inspired by image segmentation [26], edge-based PCS methods efficiently handle ideal point cloud data with low noise and uniform density. Region-growing PCS methods start by selecting seed points or units as single points [5], [31], [32], region units [33], [34], or hybrid units. Model fitting methods match point cloud data to primitive geometric shapes (planes, spheres, and cylinders) using techniques such as Hough transform (HT) [35], [37] and random sample consensus (RANSAC) [38]. Although the above-described methods made progress in single-wavelength LiDAR PCS, most of them are unsuitable for MPC data due to ignoring the use of spectral signatures.

Supervoxel segmentation is a highly versatile unsupervised method widely utilized in RGB-D point cloud data (a type of MPC in computer vision) due to its ability to efficiently incorporate multiband intensity information [39], [40], [41], [42]. The main idea is to oversegment a raw point cloud into small regions (supervoxel) to feed the postprocessing PCS algorithms rather than initial points, reducing calculation costs and the effects of noise. For example, a voxel cloud connectivity segmentation (VCCS) method [39] uses the octree voxel point cloud and employs the K-means clustering algorithm to realize supervoxel segmentation. However, the original VCCS adopts fixed resolution and relies on the initialization of seed points, and the result appears as uniformly distributed rectangles, violates the real feature boundaries, and overdependence on uniformed seed point selection. Lately, Song et al. [43] developed a boundary-enhanced supervoxel segmentation method using a two-stage oversegmentation approach that maintains object shape but assumes that points are ordered sequentially in one direction. Lin et al. [26] formalized oversegmentation as a subset selection problem and addressed the limitations of previous

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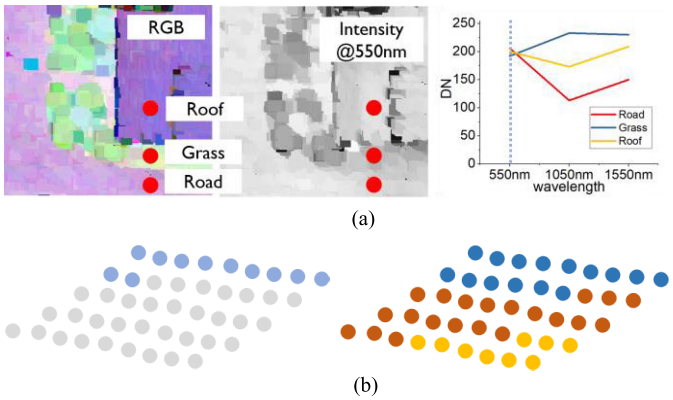
(a) (b) (c)

Fig. 1. Certain parameters segmentation method may cause undersegmentation and mis-segmentation. (a) Correct PCS in a stereoscopic view, different voxels marked as different colors. (b) Random seed point selection may result in undersegmentation (seed points indicated as bigger size). (c) Uneven scan lines caused by ALS external factors, leading to false PCS result with fixed seed resolution.

studies by adopting an adaptive resolution to preserve boundaries. Sha et al. [44] utilized a variety of searching methods to obtain neighborhoods and improved the clustering results by directly applying the local k-means clustering technique to points, rather than voxels. Landrieu and Boussaha [45] presented the first supervised framework for 3-D point cloud oversegmentation, achieving significant improvements compared to [26] and [39]. For PCS tasks, several studies have explored supervoxel-based presegmentation [46], [47], [48], [49], [50], [51]. Yao et al. [52] used mean a shift to oversegment ALS data in urban areas, which can also be employed in presegmentation.

Although the above-described supervoxel-based methods demonstrated success in PCS, two important challenges remain. First, the abovementioned methods are designed with uniformly distributed primitives for irregular MPC, resulting in an undersegmentation problem. An illustration is shown in Fig. 1(b), and these methods faced with the error from structures smaller than a fixed resolution of seed points will be covered by some voxels compared to Fig. 1(a). Lin et al. [26] formalized oversegmentation as a subset selection problem and addressed the limitations of previous studies by adopting an adaptive resolution to preserve boundaries. Second, current methods are often limited to single-wavelength LiDAR data and do not consider spectral information. The spectral curves of different objects may show similar intensities in a single band, as shown in Fig. 2(a); roof, ground, and grass share similar spatial characteristics and exhibit comparable intensities around 550 nm. Consequently, segmentation algorithms that solely rely on single-band intensity information are susceptible to causing mis-segmentation issues. A nonlinear distance metric proposed in [44] avoids the need for computations to convert from RGB space to LUV space. Segmenting the different features in a voxel labeled with majority voting causes undersegmentation and lowers the performance ceiling [49]. Meanwhile, the spatial geometric-based methods in ALS data must face the problem of uneven point cloud density caused by external factors as the different routes, the distance of scanning line in the same feature may be much larger than the distance between different features, and the geometric-based methods possibly cause mis-segmentation of the MS-LiDAR data like Fig. 1(c), which is still a significant improvement direction.

The focus of this article is to take advantage of the rich spectral information of MPC data by clustering the points without manual annotation. Our method consists of three parts: 1) normalized spectral–spatial metric segmentation; 2) energy minimization voxel merging; and 3) concave–convex feature edge optimization. The proposed method starts with a normalized spectral–spatial metric to accurately cluster points into voxels and then selects seed points without any preset parameters through an energy minimization model, which also helps to regularize the boundaries and reduce undersegmentation. Finally, the edges are optimized and adjacent voxels are merged based on concave–convex features judgment, reducing mis-segmentation errors caused by voxels crossing boundaries.



(a)

(b)

Fig. 2. Utilizing spectral information from MPC data can reduce mis-segmentation issues caused by the similarity intensity of different objects in single-band PC data. (a) Spectral of TITAN data (point size = 20) in the multispectral data (left, R = 1550nm, G = 1050 nm, and B = 550 nm) compared with single band (right, intensity@550 nm) at the same point. (b) Undersegmentation result (left) and the truth of ground (right).

The main contributions of this article are given as follows.

1) A normalized spectral–spatial metric is proposed for MPC data segmentation. Its main advantage is to cluster MPC accurately without many predefined supervoxel parameters in an unsupervised way. This metric can be applied to the hyperspectral point cloud data.
2) An energy minimization principle based on spectral and spatial characteristics is introduced for seed point selection. Compared with current supervoxel segmentation in uniform voxels, it reduces the undersegmentation error (UE) effectively and avoids manual tuning of the parameters.
3) A boundary optimization method with concave and convex judgment is devised to determine the relationship between the adjacent voxels before reallocating boundaries with spectral–spatial weights. It obviously reduces the errors from the voxels crossing the boundaries and correctly regularizes the boundaries.

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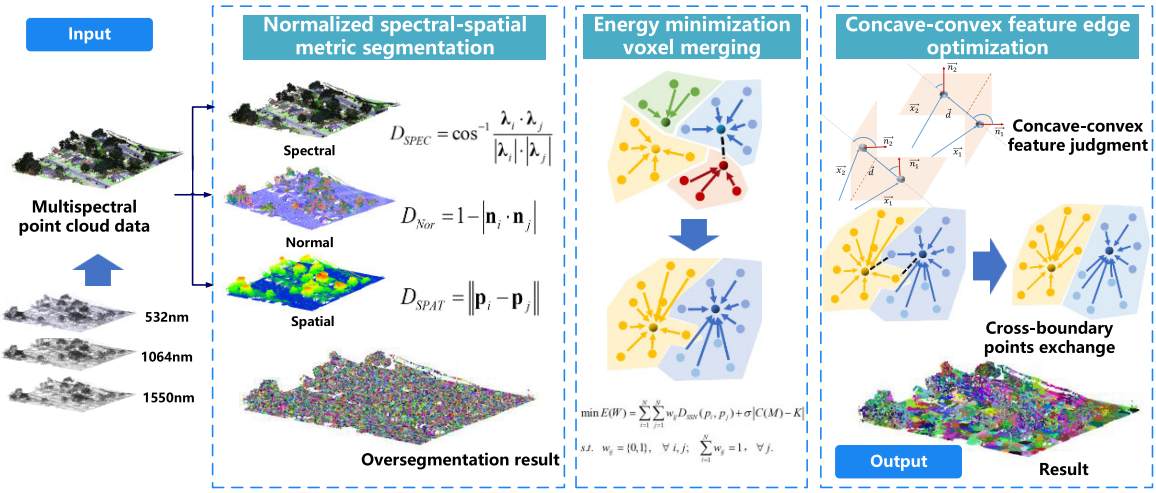


Fig. 3. Overall architecture of the proposed method, including normalized spectral–spatial metric segmentation, energy minimization voxel merging, and concave–convex feature edge optimization.

The rest of this article is organized as follows. Section II describes the proposed segmentation method for the multispectral LiDAR dataset in detail, and presents the color similarity weight designed to construct the supervoxel point clouds and the segmentation method in spectral. The experimental results and analysis are reported in Section III. Section IV concludes this article.

II. Normalized Spectral–Spatial Supervoxel Segmentation

To take advantage of the rich spectral information and address the mis-segmentation errors of airborne MPC data segmentation, a normalized spectral–spatial segmentation method is presented for airborne MPC data, clustering the points without relying on manual annotation. The overall architecture of the proposed method is shown in Fig. 3. It consists of three main components: 1) normalized spectral–spatial metric segmentation; 2) energy minimization voxel merging; and 3) concave–convex feature edge optimization. The details of the three components are presented in the following. To enhance readability, the definitions of the symbols used in this article are presented in Table I.

A. Normalized Spectral–Spatial Metric Segmentation

The goal of multispectral PCS (MPCS) is to subset the points into small units with the same class. As adjacent points with high spectral–spatial consistency [21], inspired by the supervoxel segmentation [39] in computer vision, a normalized spectral–spatial metric segmentation for MPC is proposed to make use of spectral and spatial information.

MPC data contain rich spectral information, each point has independent coordinate and spectral–spatial characteristics, and record as $\mathbf{p} = \{x, y, z, \lambda_1, \ldots, \lambda_b\}$, where $\mathbf{c} = \{x, y, z\}$ refer to the coordinate and $\lambda_p = \{\lambda_1, \ldots, \lambda_b\}$ refer to the spectral vector of the point p, with b the number of spectral bands. We express MPC structurally with k-dimensional tree (KD tree) and cluster the points with the spectral–spatial information. MPC data can be set as $\mathbf{P} = \{\mathbf{p}_1, \ldots, \mathbf{p}_N\}$ with N points, partitioned into K supervoxels, $V = \{V_1, \ldots, V_K\}$. The points are mapped into K supervoxels and labeled $\{1, \ldots, K\}$. $v{:}\{\mathbf{p}_1, \ldots, \mathbf{p}_N\} \rightarrow \{1, \ldots, K\}$, point p labeled as $v(p)$, and the voxel V_K is defined a set of points labeled $k{:}V_k = \{p|v(p) = k\}$.

D_{Spec}, D_{Spat}, and D_{Nor} are used to evaluate the distance D_{SSN} between points and build the KD tree. λ, μ, and ε are the weights of D_{Spec}, D_{Spat}, and D_{Nor}

$$D_{\text{SSN}} = \sqrt{\lambda D_{\text{Spec}}^2 + \mu D_{\text{Spat}}^2 + \varepsilon D_{\text{Nor}}^2}. \tag{1}$$

Spectral angle mapping (SAM) is adopted to build spectral similarity between the points as

$$D_{\text{Spec}} = \cos^{-1}\frac{\lambda_i \cdot \lambda_j}{|\lambda_i| \cdot |\lambda_j|} = \cos^{-1}\frac{\sum_{i=1}^{N}\sum_{j=1}^{N}\lambda_i\lambda_j}{\sqrt{\sum_{i=1}^{N}\lambda_i^2}\sqrt{\sum_{j=1}^{N}\lambda_j^2}} \tag{2}$$

λ_i and λ_j are the spectral of points p_i and p_j, $D_{\text{Spec}} \in [0, \pi/2]$. The Euclidean distance is used to denote the spatial similarity D_{Spat} between points in coordinate space. Since normal vector is an important feature of MPC data, D_{Nor} is calculated in fast point feature histogram (FPFH) space

$$D_{\text{Spat}} = \|\mathbf{c}_i - \mathbf{c}_j\|, \quad D_{\text{Spat}} \in [0, R] \tag{3}$$

$$D_{\text{Nor}} = 1 - |\mathbf{n}_i \cdot \mathbf{n}_j|, \quad D_{\text{Nor}} \in [0, 2]. \tag{4}$$

c_i and c_j are the spatial vector and $\mathbf{n}_i$ and $\mathbf{n}_j$ are the normal vector of i and j. The measure is adopted similar to VCCS, and differently, VCCS faces the point cloud data from RGB-D; they evaluate the color distance D_C in the CIELab space [39]. Lin et al. [26] (written as TBBS) set the metric D_{TBBS} without spectral information and only used spatial and vector characteristics as (5), and R_{Seed} is the resolution of the initial point set

$$D_{\text{TBBS}} = 1 - |\mathbf{n}_i \cdot \mathbf{n}_j| + 0.4 \times \frac{\|\mathbf{c}_i - \mathbf{c}_j\|^2}{R_{\text{seed}}^2}. \tag{5}$$

As each distance factor has a different range of values, the normalization approach is required when evaluating D_{SSN}

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TABLE I
SUMMARY OF MAIN SYMBOLS

Symbol	Meaning	Relationship to related symbols
$\mathbf{P}$	Multispectral point cloud data of ALS	
p	Point of MPC	$\mathbf{P}=\{\mathbf{p}_1,...,\mathbf{p}_N\}$,
$\mathbf{p}_i$	Vector of point i with spectral and spatial information	$\mathbf{p}_i=\{x_i,y_i,z_i,\lambda_1,...,\lambda_b\}$,
$\boldsymbol{\lambda}_i$	Spectral information of point $\mathbf{p}_i$ with b bands	$\mathbf{p}_i=\{\mathbf{c}_i,\boldsymbol{\lambda}_i\}$, $\boldsymbol{\lambda}_i=\{\lambda_1,...,\lambda_b\}$, $\mathbf{c}_i=\{x_i,y_i,z_i\}$
$\mathbf{c}_i$	Coordinate of point $\mathbf{p}_i$	
V	Segmentation result	$V=\{V_1,...,V_K\}$, $v:\{\mathbf{p}_1,...,\mathbf{p}_N\}\rightarrow\{1,...,K\}$
$v(p)$	Voxel p of segmentation result in V	
$\mathbf{N}$	Surface normal	$\mathbf{N}=\{\mathbf{n}_1,\mathbf{n}_2,\mathbf{n}_3...\mathbf{n}_i\}\in\mathbf{P}$
$\mathbf{n}_i$	Surface normal on point i	
S_{single}	the number of voxels consisting of points labeled with the same tags	$S_{single}\leq S_{all}$
S_{all}	total number of voxels	

between points for normalization. These distances need to be normalized with λ, μ, and ε, according to the range of D_{Spec}, D_{Spat}, and D_{Nor}; the weights are set as follows and R is the resolution of supervoxels

$$\lambda=\frac{4}{\pi^2},\quad \mu=\frac{1}{R},\quad \varepsilon=\frac{1}{2}. \tag{6}$$

B. Energy Minimization Voxel Merging

Traditional supervoxel segmentation methods often employ uniform voxels ignoring the irregular data characteristics, resulting in strict cubic voxel distributions against the actual structure and distribution of features, which leads to mis-segmentation in Fig. 1(b). The minimized distance model with D_{SSN} is used for representative point selection adaptively rather than presetting the measurements of voxels in VCCS. This model can merge the adjacent voxels with similar features and reduce the number of voxels, which is regarded as an energy minimization principle.

The representative point of voxel V_i is recorded as r_i, which is the root node of the voxel; record the unrepresentative point in the same voxel as p_i; $C(M)$ is the count of representative points r_i of MPC; and the mapping function can be written as $v(p)=\arg\min_i D_{\text{SSN}}(p,V_i)$. Each point is represented exactly in a single voxel, assuming to have already obtained K voxels and calculate the sum distance of $\{V_1,\ldots,V_K\}$ as $\sum_{i=1}^{N}\sum_{j=1}^{N}D_{\text{SSN}}(p_i,p_j)$. Here, a binary variable $w_{ij}\in\{0,1\}$ is used to encode the subset selection problem. $w_{ij}=0$ means that p_i is not the representative points of p_j

$$E(W)=\sum_{i=1}^{N}\sum_{j=1}^{N}w_{ij}D_{\text{SSN}}\left(p_i,p_j\right). \tag{7}$$

To find the optional solution, K is used to limit $C(M)$ and use σ to weight the tradeoff as the regularization parameter, and the minimization distance model can be formalized as

$$\begin{aligned}&\min\ E(W)=\sum_{i=1}^{N}\sum_{j=1}^{N}w_{ij}D_{\text{SSN}}\left(p_i,p_j\right)+\sigma|C(M)-K|\\&\text{s.t. } w_{ij}=\{0,1\}\quad\forall i,j;\quad\sum_{i=1}^{N}w_{ij}=1\quad\forall j.\end{aligned} \tag{8}$$

Initialize σ with the median of minimum D_{SSN} between points and double for iteration quickly

$$\sigma_0=\text{median}\left\{\min\left(D_{\text{SSN}}\left(p_i,p_j\right)\right)\right\}. \tag{9}$$

The whole procedure is summarized in Algorithm 1

Algorithm 1 Energy Minimization Voxel Merging

Input: A point set, $\boldsymbol{P}$; the neighboring set, $\boldsymbol{N}$; the dissimilarity metric, D_{SSN}; and the expected number of supervoxels, K.
Output: K supervoxels $\{S_1,\ldots,S_k\}$
Initialize $\sigma_0=median\{\min_p(D_{\text{SSN}}(p,\boldsymbol{N}_i))|p\in\boldsymbol{P}\}$
$\sigma\leftarrow\sigma_0$; $G\leftarrow\boldsymbol{N}$; $\boldsymbol{R}\leftarrow\boldsymbol{P}$; $V_i\leftarrow 1$
repeat
  for all $r_i\in\boldsymbol{R}$ **do**
    for all $r_j\in G_i\cap\boldsymbol{R}$ **do**
      if $\sigma-c_jD_{\text{SSN}}(r_j,r_i)>0$ **then**
        Merge r_j into r_i
        $G_i\leftarrow G_i\cup G_j$
        $\boldsymbol{R}\leftarrow\boldsymbol{R}-\{r_j\}$
        $c_i\leftarrow c_i+1$
      end if
    end for
  end for
  $\sigma\leftarrow 2\sigma$
until $|\boldsymbol{R}|=K$
return $\{S_1,\ldots,S_k\}$ according to $S_i=\{p_j|w_{ij}=1\}$

C. Concave–Convex Feature Edge Optimization

While voxel merging can reduce UEs, the existence of unrepresented points with incorrect labels, as shown by cross-boundary points (among orange voxel cross the boundary) in Fig. 4(a), can result in irregular boundaries between adjacent voxels. There is still room for improvement in regard to the cross-boundary phenomenon present in unrepresentative points. The boundaries of voxels are optimized by changing unrepresentative points to improve the edges and address false segmentations.

Representing a significant feature of airborne MPC data, the boundaries of points within the same features are

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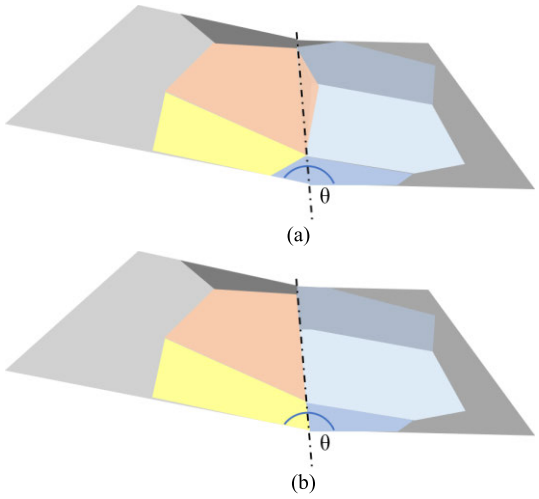


Fig. 4. Cross-boundary points still exist after voxel merging, which will make result in undersegmentation. (a) Undersegmentation voxels with cross-boundary points before edge optimization. (b) Correct segmentation result after exchange to expect.

often convex connections [49], while those between different features are concave [49]. Before any modification, it is necessary to determine the relationship between adjacent unrepresentative points to judge whether they belong to the same feature. This can be done by using normal vectors of nonrepresentative points in neighboring voxels to assess the spatial relationship, swapping cross-boundary points that conform to convex connections for edge optimization while preserving the concave edges that contain real boundary details, thus improving the segmentation accuracy.

Using the concave–convex features judgment based on the normal vectors of unrepresentative points, Wang et al. [54] used the convex connections to build the LCCP algorithm. The connection between adjacent voxels can be represented in Fig. 5. $\overrightarrow{x_1}$ and $\overrightarrow{x_2}$ are the position of the adjacent points p_1 and p_2; based on the vector $\overrightarrow{d} = \overrightarrow{x_2} - \overrightarrow{x_1}$, $\angle\alpha_1$ and $\angle\alpha_2$ are the angle of vector $\overrightarrow{d}$ with normal vectors $\overrightarrow{n_1}$ and $\overrightarrow{n_2}$, whether the connection between these is convex or concave can be inferred from the relation of $\angle\alpha_1$ and $\angle\alpha_2$.

For convex connections, α_2 is smaller than α_1. This can be expressed as

$$\alpha_1 > \alpha_2 \Rightarrow \cos(\alpha_1) - \cos(\alpha_2) < 0 \Leftrightarrow \overrightarrow{n_1} \cdot \widehat{d} - \overrightarrow{n_2} \cdot \widehat{d} < 0 \tag{10}$$

$$\widehat{d} = \frac{\overrightarrow{x_2} - \overrightarrow{x_1}}{\left\| \overrightarrow{x_2} - \overrightarrow{x_1} \right\|}. \tag{11}$$

Also, it can express the concave connections as

$$\alpha_1 < \alpha_2 \Leftrightarrow \overrightarrow{n_1} \cdot \widehat{d} - \overrightarrow{n_2} \cdot \widehat{d} > 0. \tag{12}$$

If $\alpha_1 = \alpha_2$, the points p_1 and p_2 are defined at the same plane. The convex–concave feature judgment (CC judgment) can be defined as

$$\mathrm{CC}(p_i, p_j) := \begin{cases} \alpha_j \le \alpha_i, & \text{convex} \\ \alpha_j > \alpha_i, & \text{concave}. \end{cases} \tag{13}$$

Also, p_1 and p_2 under convex connection can be calculated for exchange; otherwise, the points cannot exchange. This preserves more boundary detail. After the concave and convex connection judgment, the distance of point to each representative point is compared with the adjacent voxels. Considering the CC judgment contained the information of normal vectors, the metric D_E is defined from the point p_i to the representative point r_i and r_j with the normalized spectral and spatial information

$$D_E = \sqrt{\frac{4D_{\mathrm{Spec}}^2}{\pi^2} + \frac{D_{\mathrm{Spat}}^2}{R_{\mathrm{seed}}^2}}. \tag{14}$$

The whole procedure is summarized in Algorithm 2.

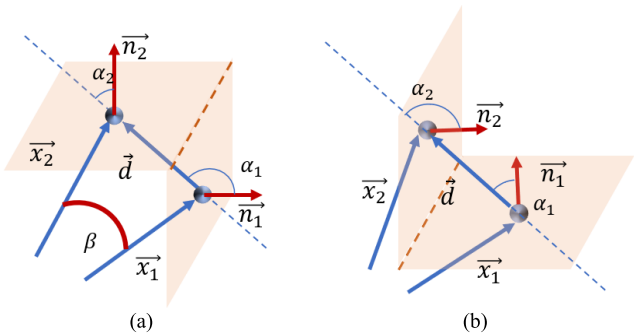


Fig. 5. Connections between convex and concave. (a) Convex connection. (b) Concave connection.

Algorithm 2 Concave–Convex Feature Edge Optimization

```
Initialize a queue, Q, for points, i.e., Q ← P
while Q ≠ ∅ do
  If α_j ≤ α_i → convex connections
    remove the front point p_j from Q
    for all p_i ∈ N_i do
      if D_E(p_i, r_j) < D_E(p_i, r_i) then
        Assign p_i to r_j
        if p_i ∉ Q then
          Add p_i to the back of Q
        end if
      end if
    end for
  else α_j > α_i → concave connections
end while
```

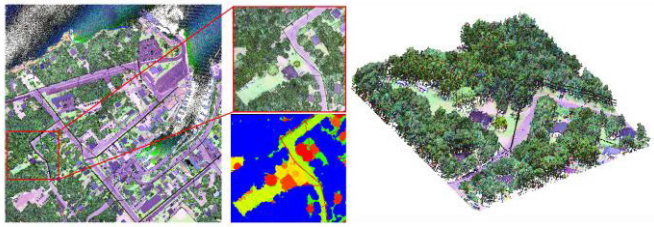
III. Experiments and Analysis

A. Data Description and Evaluation Metrics

Two datasets gained by Titan onboard system are used in the experiments. The Titan multispectral LiDAR is manufactured by Teledyne Optech Inc., and the wavelengths of channels are set at 1550-, 1064-, and 532-nm wavelengths simultaneously through a single oscillating mirror scanner at pulse repetition frequencies (PRF) that range from 50 to 300 kHz per wavelength (max combined PRF of 900 kHz). The University of Houston dataset is a public benchmark that can be downloaded from the 2018 IEEE GRSS Data Fusion Contest,[1]

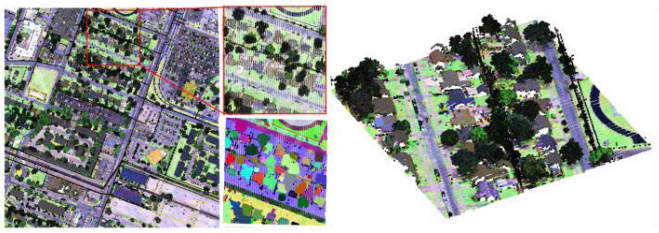
[1]https://site.ieee.org/france-grss/2018/01/16/data-fusion-contest-2018-contest-open/

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(a) (b) (c)

Fig. 6. Colored MPC and labels of Harbor of Tobermory dataset. (a) False-color map of point cloud and experimental area. (b) Experimental area and labels. (c) False color in 3-D viewer of the experimental area.



(a) (b) (c)

Fig. 7. Colored MPC and labels of University of Houston dataset. (a) False-color map of point cloud and experimental area. (b) Experimental area and labels. (c) False color in 3-D viewer of the experimental area.

and the Harbor of Tobermory dataset was obtained by emailing the competition organizer, the sensor and related parameters are consistent with the University of Houston dataset. These data are widely used in research for segmentation and classification purposes [2], [21], [55], [56]. The ground truth for supervoxel segmentation separates different objects individually based on their boundaries, localized data patches were manually labeled to assess the accuracy of supervoxel segmentation, and the ground truth for the classification results of the entire dataset was provided by Wang et al. [21].

Each channel of Titan system is individual, and the returned laser pulse has the intensity value at the corresponding wavelength. In order to use the point cloud with multispectral information, we synthesize the raw data of Titan into single-band point cloud; 1550 nm is used as a basis because water has the strongest absorption of laser in this band, which has the least number of corresponding point clouds. By searching the nearest neighbor point of 1550 nm in spatial, the multispectral point cloud (MPC) used in this article is generated by the value of the nearest neighbor point in 1064 and 532 nm and merges the point cloud as: $[x, y, z, \mathrm{LRI}_{\lambda_{1550}}, \mathrm{LRI}_{\lambda_{1064}}, \mathrm{LRI}_{\lambda_{532}}]$. The Harbor of Tobermory and the University of Houston datasets are used in this article.

Harbor of Tobermory: This dataset was collected from a coastal port area in Tobermory, Canada. Different from point cloud classification, with the purpose of segmentation, objects need to be labeled separately instead of in a certain class. Our experimental area, a cabin in the woods with 7 36 302 points, was selected as a representative feature in 140 m × 130 m and labeled as 50 distinct objects, including forests, cottages, roads, powerlines, cars, and grass. The false-color LiDAR data and labels can be seen in Fig. 6, and the truth is shown in Fig. 6(b).

University of Houston: This dataset gained from the University of Houston; the experimental area is a villa area, and it has 2 98 683 points and is labeled as 183 distinct objects, including land, grass, road, trees, powerlines, cars, lamp, cottages, and bush. The false-color point cloud and labels are shown in Fig. 7, and the truth of segmentation is shown in Fig. 7(b).

Two standard indexes are used, which were commonly used to evaluate the accuracy of the segmentation and propose an index to describe the voxels' purity.

1) Undersegmentation Error (UE): The UE is used to evaluate the segmentation result. It is an important metric for calculating the points number of supervoxels that across on boundaries. We adopt the same definition of UE given in [39]

$$\mathrm{UE} = \frac{1}{N}\left(\sum_{i=1}^{M}\left(\sum_{S_j|S_j\cap G_i\neq\emptyset}\left|S_j\right|\right) - N\right). \tag{15}$$

The regions of ground-truth segmentation are written as $G_1, G_2, \ldots, G_M$, and M is the number of regions. N is the total number of points of data, the supervoxels in the ground-truth G_i are determined as $\{S_1, S_2, \ldots, S_j\}$, and S also contains the voxels overlapped with the boundaries. The points of beyond the ground truth are counted and normalized by N, where the more the UE lower, the less the supervoxels cross the boundaries.

2) Pure Voxel Rate (PVR): In order to evaluate the metric influence for segmentation and edge estimation, the pure supervoxels' proportion of the segmentation results is calculated. The pure supervoxel points have a single label that corresponds to the ground truth. S_{single} refers to the number of voxels consisting of points labeled with the same tags, S_{all} refers to the total number of voxels

$$\mathrm{PVR}_{S_i} = \frac{\sum_i^P S_{\mathrm{single}}}{\sum S_{\mathrm{all}}}. \tag{16}$$

3) Feature Classification Error (FCE): For MPC interpretation tasks, the FCE is used. FCE measures the spectral similarity of points within one voxel. The formulated description is given as follows:

$$\mathrm{FCE} = \frac{1}{N}\left(\sum_{i}^{M}\left(\sum_{p_j|G(p_j)=G(S_i),p_j\in S_i}\left|p_j\right|\right)\right). \tag{17}$$

The regions of ground-truth segmentation are written as $G(S_i)$, and the point j is written as p_j.

In order to validate the effectiveness of the proposed method, the experiments are designed to evaluate the results of supervoxel segmentation and merging of the multispectral LiDAR data. For comparison, we implemented the method according to the public source code in the point cloud library (PCL). First, TBBS [26] is improved by using normalized spectral–spatial information. We retrieved the author's code from GitHub, and the same parameters are set for comparison, and in order to find out our method improvement, to obtain a more intuitive segmentation result, the resolution is set as $R = 6.0$ and followed the settings of the normal and spatial

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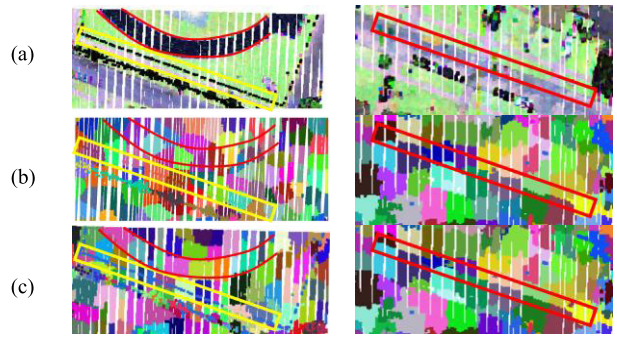


Fig. 8. For the local data of the University of Houston dataset segmentation result of normalized spectral–spatial metric improvement, $R = 6.0$ is set to obtain a more intuitive visual. (a) Local scene. (b) TBBS. (c) Improved.

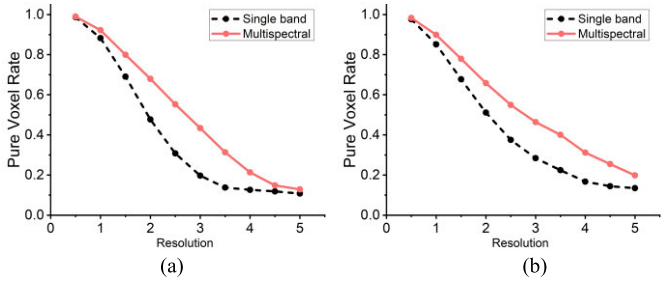


Fig. 9. PVR evaluation with single band and multispectral point cloud data of the experimental area of each dataset. (a) Harbor of Tobermory dataset. (b) University of Houston dataset.

parameter as 1 and 0.4 [26]:

$$D = 1 - \left|\mathbf{n}_i \cdot \mathbf{n}_j\right| + 0.4 \times \frac{D_{\text{Spat}}^2}{R_{\text{seed}}^2}. \tag{18}$$

Second, to find out the impact of the concavity criterion on the accuracy of segmentation results during edge optimization with adjacent voxels exchanging unrepresentative points, the results are compared after concave and convex judgment. The resolution R selected at 0.4 intervals from 1.0 to 5.8.

Then, the segmentation algorithms are compared with VCCS, VCCS-KNN, TBBS [26], VCCS-LUV that inspired from the LUV color space [44], and Nyström-based spectral clustering (NSC) [36]. For a more equitable comparison, we modified VCCS after testing the VCCS and VCCS-KNN algorithm parameters iteratively for these datasets and selected the best parameter settings as comparative validation. The results are analyzed with several indicators. The segmentation effect is evaluated by UE, PVR, and FCE. The number of neighbors is set as 100. The resolution R was selected at 0.5 intervals from 1.0 to 5.0. Finally, supervoxel segmentation can play a critical role in preprocessing, and the application aims to reduce computational workload by compressing point cloud data. It achieves this by using a single point to represent all points within a supervoxel as input, and the features are obtained through voting or averaging methods, which results in some information loss due to the resampling mechanism. The 3SGCN [21] classification algorithm relies on the results of supervoxel segmentation for training. Therefore, it becomes crucial to ensure the purity of supervoxel features, as less feature loss will occur during the voting process. Consequently, obtaining purer supervoxel

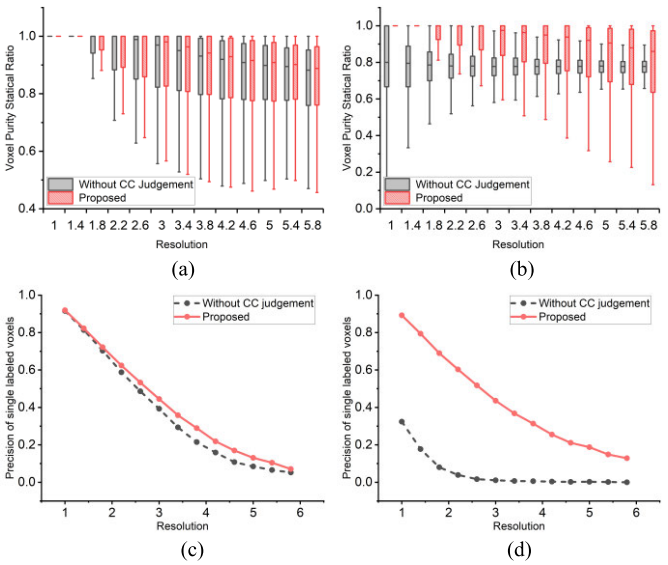


Fig. 10. Results of CC judgment in boundary regulation. (a) Statistical result of Harbor of Tobermory dataset. (b) Statistical result of University of Houston dataset. (c) PVR of the Harbor of Tobermory dataset. (d) PVR of the University of Houston dataset.

data directly influences the accuracy and reliability of the classification results in the research context. As a result, we proceeded to classify the segmentation results, and the output MPC segmentation results were utilized as inputs for the point cloud classification, thereby evaluating the influence of segmentation accuracy in practical applications. The parameters, such as overall accuracy (OA), average accuracy (AA), and Kappa, were used to assess how the accuracy of the supervoxel segmentation algorithm impacts the classification results. As the classification preprocess, the superpoints obtained through voting in smallest training units were used for training and classification, where each superpoint is represented as $\mathbf{p}_c = \{x_c, y_c, z_c, \lambda_r, \ldots, \lambda_b\}$ and its label is determined by the internal voting results of the supervoxels. We set $R = 1$ based on comparative tests to achieve the optimal performance. To ensure fairness in the evaluation, the number of supervoxels for the MPC dataset in both scenes was controlled to be around 10% of each dataset.

In all experiments, the parameters in these compared methods consist of the corresponding reference. The proposed method was coded in C++ and run on one core of an Intel Core i5-10400 2.90-GHz CPU, with 32-GB memory, on a Windows 10 system.

B. Improvement Analysis

The validity is verified by the improved part of the algorithm first. Fig. 8 visualizes the segmentation results of TBBS and the improved method with normalized spectral–spatial metric. To observe the differences in the improvement of the spectral–spatial base method visually, the voxel resolution is set as $R = 6$. It can be seen that the improved method can split the boundary more clearly, especially the boundary of the road and grass. The boundary between the road and grass is marked with the red line and the fence with yellow box

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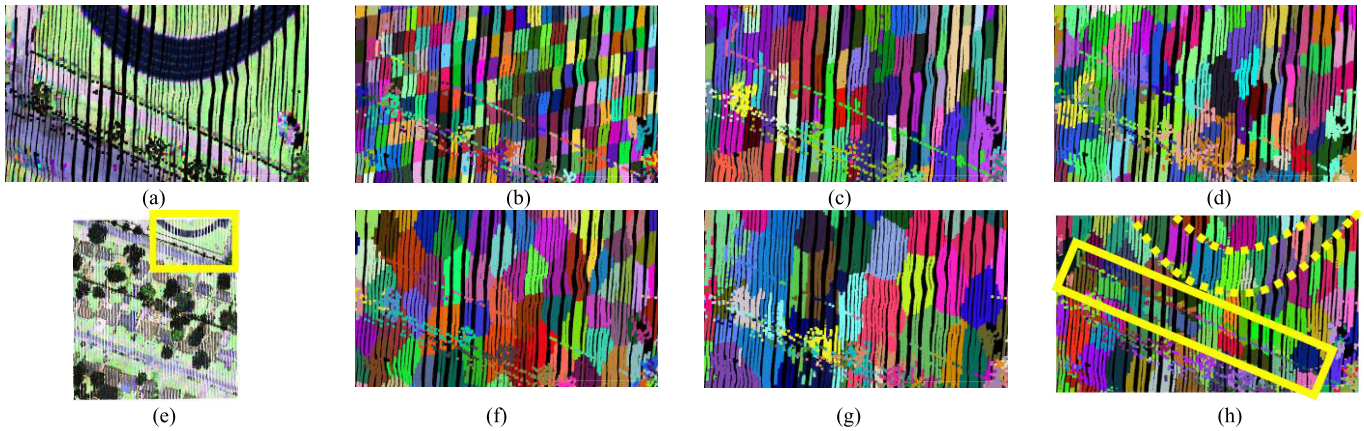


Fig. 11. Performance of the six methods in local magnification results of the University of Houston dataset, the proposed method can achieve reasonable segmentation for runway edges and roads. Even in the case of $R = 6$, the boundaries remain neat and distinct. (a) MPC data. (b) VCCS. (c) VCCS_KNN. (d) TBBS. (e) Location. (f) VCCS_LUV. (g) NSC. (h) Proposed.

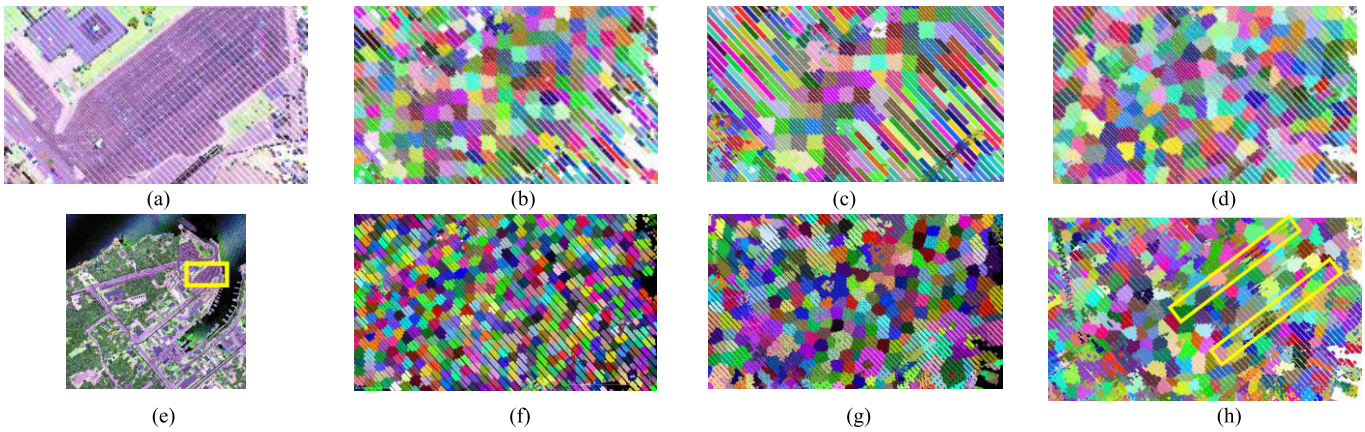


Fig. 12. Performance of the six methods in local magnification results of Harbor of Tobermory dataset, the proposed method can effectively segment the white boundaries of the parking lot ground which is highlighted with yellow bounding boxes, allowing us to clearly observe the parallel distribution of the supervoxel segmentation result. (a) MPC data. (b) VCCS. (c) VCCS_KNN. (d) TBBS. (e) Location. (f) VCCS_LUV. (g) NSC. (h) Proposed.

line. The metric proposed can get more accurate boundaries of point cloud data. PVR is also used to evaluate both datasets with experiment and the resolution R selected at 0.5 intervals from 0.5 to 5.0. Fig. 9 reports that the spectral–spatial-based metric has a better performance, and multispectral information can subset the points better than single-band intensity.

The proposed method's statistical results in edge optimized compared with the merging method without CC judgment of the experimental datasets separately shown in Fig. 10(a) and (b) show the statistic of segmentation result about each voxel's pure rate.

In these diagrams, the middle line represents the median, the boxes represent the interquartile range (IQR), which shows the upper and lower quartiles of statistical results, and the ends of the lines represent the range of upper and lower bounds of the statistical results. The rule of outliers is set to 1.5. The boxplot shows that the precision gets lower with the increasing R and the proposed method has a better result, which uses the CC judgment before exchanging the cross boundaries points. In the Harbor of Tobermory dataset, Fig. 10(a) shows the IQR of each result rising gradually. The lower quartiles of both methods at $R = 1.0$ and $R = 1.4$ are 1 but decrease quickly; the median starts to lower at $R = 2.6$ without CC judgment method and the median begins to lower at $R = 3.0$ of the proposed method, which means that the proposed method has a better result. In the University of Houston dataset, Fig. 10(b) shows that the lower quartiles of proposed method at $R = 1.0$ and $R = 1.4$ are 1 and the median starts to lower at $R = 3.0$. The median of the method without CC judgment is floated at a value of 0.8, meaning that our method can optimize the edge better. Fig. 10(c) and (d) shows that the proposed method has higher PVR results at corresponding R, and the higher PVR value means that our method can gain more pure voxels with the CC judgment. By employing CC judgment, our approach can yield purer supervoxels and a smaller resolution R will get more pure voxels but also gain a larger number of smaller voxels, which will limit the efficiency of our operations. Figs. 11 and 12 visualize the local details of the result, Fig. 12 shows a parking lot with yellow lines of Harbor of Tobermory dataset, and our method can separate parking lines with D_{spec} from parking lot circled by red wireframes.

C. Experimental Results and Analysis

Some typical results on semantic 3-D benchmark are shown in Figs. 11 and 12. The result has the neat boundary and high similarity with the ground truth and the spectral-based segmentation method can separate the road and grass

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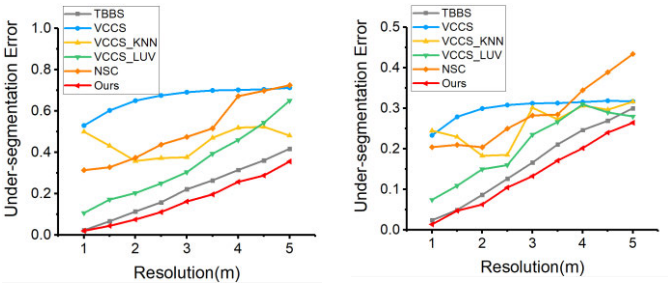


Fig. 13. UE evaluation of four methods. (a) University of Houston dataset. (b) Harbor of Tobermory dataset.

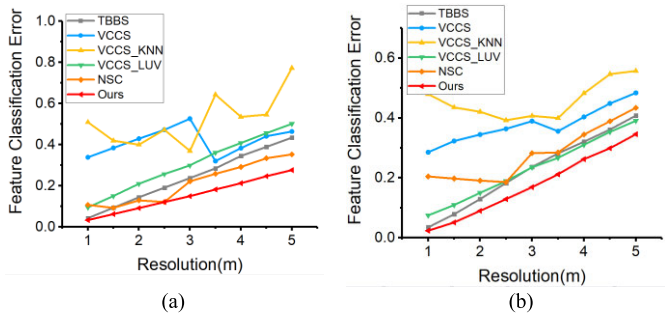


Fig. 14. FCE evaluation of four methods. (a) University of Houston dataset. (b) Harbor of Tobermory dataset.

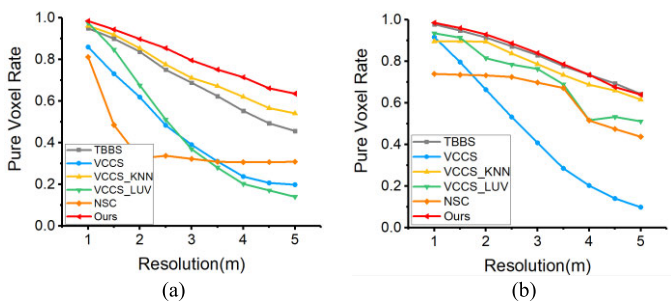


Fig. 15. PVR evaluation of four methods. (a) University of Houston dataset. (b) Harbor of Tobermory dataset.

TABLE II
CLASSIFICATION RESULTS OF VARIOUS SEGMENTATION METHODS ON HARBOR OF TOBERMORY DATASET (%)

Method	VCCS	VCCS_KNN	TBBS	VCCS_LUV	NSC	Ours
OA	85.85	85.85	91.39	90.34	87.12	**92.97**
AA	86.08	87.43	84.60	**90.68**	85.35	89.77
Kappa	83.27	84.09	90.9	89.13	85.51	**92.09**

TABLE III
CLASSIFICATION RESULTS OF VARIOUS SEGMENTATION METHODS ON UNIVERSITY OF HOUSTON DATASET (%)

Method	VCCS	VCCS_KNN	TBBS	VCCS_LUV	NSC	Ours
OA	85.28	88.60	92.62	85.70	76.97	**93.21**
AA	86.97	90.02	92.51	87.36	65.94	**96.30**
Kappa	83.18	86.97	91.58	83.67	73.69	**92.24**

obviously. The results shown in Fig. 11 is not the optimal value. However, it serves the purpose of visually demonstrating the effectiveness of the segmentation method proposed in this study based on spatial–spectral characteristics at a larger parameter $R = 3$. The proposed method shows improved accuracy in boundary segmentation, particularly for areas with similar intensities at 550 nm, such as grass and road. Our results indicate that the proposed method successfully distinguishes track, grassland, and roads while also achieving satisfactory segmentation for fences and power lines.

The white part of Fig. 12(b)–(d) means the UE and labeled 0. Because the nonuniformity of LiDAR scan lines and airstrips are not parallelism, the undersegmentation results of VCCS and VCCS_KNN based on the spatial information which contain many striated voxels with highly similar to the point cloud density texture, against the actual feature types. The voxels of VCCS and VCCS_KNN based on the present size of the seed resolution and voxels are measured, which will reduce the accuracy. In Fig. 11, for the University of Houston dataset, we can see that the voxels of the proposed method are not influenced very much by the density of dataset, but VCCS and VCCS_KNN confused the road and grass and the voxels parallel with the LiDAR scan direction; TBBS cannot divide the features neatly. The proposed method splits the boundary clearly with spectral–spatial-based metric, especially the boundary of the road and grass.

As shown in Figs. 13–15, the proposed method achieves the best UE, FCE, and PVR values at all resolution settings. Intuitively, the proposed method achieves better perceptually satisfactory segmentation results.

The result of UE shows that the error increases with R during supervoxel segmentation. As the radius increases, due to the negative correlation between R and D_{SSN} in the spatial–spectral normalized metric, when D_{SSN} remains constant, the threshold range of D_{Spec} and D_{Nor} increases. Consequently, the UE also increases. In contrast, our method can better suppress the occurrence of undersegmentation. As R increases, FCE also increases, whereas PVR decreases. Clearly, as R increases, individual supervoxels will accommodate more point cloud data and undergo recombination based on energy minimization. The results of classification in Tables II and III indicate that our proposed method can achieve higher classification, implying that our method can obtain purer supervoxel results. From OA, AA, and Kappa, we can observe that our method exhibits higher classification accuracy. This suggests that, during segmentation, our supervoxels are purer compared to supervoxels obtained through other methods. In comparison to the VCCS, VCCS_KNN, TBBS, VCCS_LUV, and NCS methods, the normalized spatial–spectral segmentation method proposed in this study, based on spectral feature normalization, can achieve more accurate segmentation of MPC data. Our method provides purified preprocessed voxel data for classification applications. Notably, even with a reduction in the point cloud data to only 10% of the original size, our method maintains a higher classification accuracy.

IV. CONCLUSION

A normalized spectral–spatial unsupervised segmentation method for MPC segmentation is proposed in this article, which is based on the energy minimization voxel merge

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technique and concave–convex feature edge optimization. The method effectively clusters the points into voxels for scenes such as urban and forest, and can successfully handle the MPC that has differences in density from uneven scan lines. The proposed method also offers a unique advantage over other multispectral ALS supervoxel segmentation methods by using normalized spectral and spatial information of the MPC data rather than relying on preset voxel parameters. This results in an accurate segmentation outcome with reduced data volume, low information loss, and minimal overlap with the original point cloud data. Moreover, optimizing boundaries with the CC judgment framework greatly improves the accuracy of the voxel purity and merges the cross-border points into the right voxels to get the correct edge of features, serving as a reliable preprocessing method for MPC.

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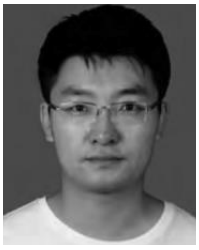
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